VU Medical Zone

BT 402 Current Final term Spring 2021

Q1. Define Microbial Mining? 2 Marks

The use of microorganisms to remove minerals particularly metals from rocks. Biomining is a technique of extracting metals from ores and other solid materials typically using prokaryotes or fungi.

Q2. Define probiotic? 2 Marks

In recent years, probiotic cultures have become popular in dairy products because of their health benefits. These cultures are all very carefully selected strains, and there is good evidence that they help improve digestion, safeguard the immune system, and keep the body's intestinal flora in balance.

Q3. How Second-generation biofuel are produced? 3 Marks

Second-generation biofuels are produced from non-food crops, such as cellulosic biofuels and waste biomass (stalks of wheat and corn, and wood). Common second-generation biofuels include vegetable oils, biodiesel, bio alcohols, biogas, solid biofuels, and syngas.

Q4. What are the criteria for Microorganism to include in probiotic group? 3 Marks

These cultures are all very carefully selected strains, and there is good evidence that they help improve digestion, safeguard the immune system, and keep the body's intestinal flora in balance.

Q5. Write down two Types of bio fertilizers? 5 Marks

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Nitrogen bio fertilizers

• The sort of bio fertilizers can help any farmers to determine the nitrogen point during the solid ground. Nitrogen may be a crucial factor which happens to be put to use in any growing for put.

• Including, Azotobacteria must be used to your non-legume bounty; Rhizobium it takes to your legume bounty. Purple earth-friendly algae are needed to progress hemp despite the fact that Acetobacter must be used to progress sugarcane.

Phosphorus bio fertilizers

Phosphorus bio fertilizers widely used to determine the phosphorus point during the solid ground. The decision for phosphorus to your put growing is small. Phosphorus bio fertilizers come up with any solid ground obtain demanded sum of phosphorus.

• Isn't crucial that your particular selected phosphorus bio fertilizers must be used for that selected types of head. They are put to use in all different kinds of any head including; Acetobacter, Rhizobium can implement.

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Q6. What are the aspects of Polysaccharide structure? 5 Marks

The glycosidic bond between two monosaccharides is formed between the hydroxyl group on the anomeric carbon of one monosaccharide and any hydroxyl group on the other monosaccharide. Hence, the formation of a disaccharide from two identical hexopyranose ring structures of the d-series may result in 11 different isomers. In eight of the isomers, the glycosidic linkage is between C-1, in either the α or β anomeric configuration, and C-2, C-3, C-4, or C-6 of the other pyranose residues. These are designated α-D- (1→2) linkages, α-D-(1→3), linkages, and so on, where α and β refer to the anomeric configuration at C-1. The other three isomers are created by acetal formation between both C-1 atoms through the glycosidic oxygen atom in the α,α, α,β, or β,β configuration. A similar series of 11 isomers will result if the two identical hexopyranose residues are of the l-series. The number of possible isomers is even higher if furanose forms (five-membered rings) are included. When the two monosaccharides are not identical, the number of possible structures increases, because either carbohydrate residue can occupy the first or the second position – that is, can be the reducing or non-reducing residue. Each additional

carbohydrate residue brings a large increase in the number of possible isomers. For complex polysaccharides, the number of theoretically possible structures is astronomical.

Q7. What are antimicrobial pesticides? 5 Marks

Antimicrobial pesticides

Antimicrobial pesticides help to slow the growth of human pathogenic microorganisms and are used in food processing plants, dairies, breweries, poultry houses and other animal feeding operations, hospitals and medical and dental clinics and offices, municipal drinking water and water treatment facilities, swimming pools and spas, schools, day-care centers, public access facilities and homes. According to the U.S. Environmental Protection Agency (EPA), and defined by the Federal Insecticide, Fungicide, and Rodenticide Act, antimicrobial pesticides are used in order to control growth of microbes through disinfection, sanitation, or reduction of development and to protect inanimate objects, industrial processes or systems, surfaces, water, or other chemical substances from contamination, fouling, or deterioration caused by bacteria, viruses, fungi, protozoa, algae, or slime.

The EPA monitors products, such as disinfectants/sanitizers for use in hospitals or homes, in order to ascertain efficacy. Products that are meant for public health are therefore under this monitoring system, including products used for drinking water, swimming pools, food sanitation, and other environmental surfaces. These pesticide products are registered under the premise that, when used properly, they do not demonstrate unreasonable side effects to humans or the environment. Even once certain products are on the market, the EPA continues to monitor and evaluate them to make sure they maintain efficacy in protecting public health.

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Q8. What is Biodiesel and how it is produced? 5 Marks

Biodiesel is produced, mainly in the European Union, by combining vegetable oil or animal fat with an alcohol. Biodiesel can be blended with traditional diesel fuel or burned in its pure form in compression ignition engines. Its energy content is somewhat less than that of diesel (88 to 95%). Biodiesel can be derived from a wide range of oils, including rapeseed, soybean, palm,

coconut or jatropha oils and therefore the resulting fuels can display a greater variety of physical properties than ethanol.

Q9. Application methods of bio fertilizers? 2 Marks

• Seedling root dip,

• This method is applied to the rice crop. A bed of water is spread on the land where the crop has to grow. The seedlings of rice are planted in the water and kept there for eight to ten hours.

Q10. How we obtained biofuel? 2 Marks

Biofuels are fuels produced directly or indirectly from organic material biomass, including plant materials and animal waste.

Q11. Benefits of Compositing? 3 Marks

• Soil conditioner: With compost, you are creating rich humus for lawn and garden. This adds nutrients to your plants and helps retain moisture in the soil.

• Recycles kitchen and yard waste: Composting can divert as much as 30% of household waste away from the garbage can.

• Introduces beneficial organisms to the soil: microscopic organisms in compost help aerate the soil, break down organic material for plant use and ward off plant disease.

• Good for the environment: Composting offers a natural alternative to chemical fertilizers.

• Reduces landfill waste: Most landfills in North America are quickly filling up; many have already closed down. One-third of landfill waste is made up of compostable materials.

Q12. How we apply Biofertilizer? 5 Marks

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Seed treatment

Within this process your nitrogen along with phosphorus fertilizers are generally put together jointly throughout normal water. And then vegetables are generally dropped within this mix. Following purposes on this substance on the vegetables, vegetables are generally dried up. As soon as they normally dry out, they must always be sown immediately ahead of that they find harmed by simply unsafe bacteria.

Soil treatment

All the bio fertilizers along with the compost fertilizers are mixed together. They are kept for one night. Then the next day the mixture is spread on the soil where seeds have to be sown.

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Q13. Differentiate between Submerged Fermentation and Solid-state fermentation? 5 Marks

Submerged Fermentations:

Submerged Fermentations are those in which the nutrient substratum is liquid and the organism grows inside the substratum. The culture conditions are made uniform with the help of spargers and impeller blades. Most of the industrial fermentations are of this type. The substratum which is in a liquid state and such medium is also called as broth.

Solid Substrate/State Fermentation:

Solid state (substratum) fermentation (SSF) is generally defined as the growth of the microorganism on moist solid materials in the absence or near the absence of free water. In recent years SSF has shown much promise in the development of several bioprocesses and products, SSF has been ambiguously used as solid-state fermentation or solid-substrate fermentation.

Q14. Define Soil Treatment? 2 Marks

Soil treatment

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All the bio fertilizers along with the compost fertilizers are mixed together. They are kept for one night. Then the next day the mixture is spread on the soil where seeds have to be sown.

Q15. Xanthan gum. 2 Marks

Xanthan gum was discovered in the mid-1950s, during a systematic search for useful biopolymers. A bacterium Xanthomonas campestris (originally isolated from the rutabaga plant) produced a polysaccharide with potentially valuable physical properties.

Q16. Interaction between Plants and Microbes? 5 Marks

In nature: microbe plant interaction

- Many microbes are fond in nature and help plants to grow
- Rhizobacteria found in the soil fixate nitrogen which is required to grow many crops

Recycling of masses

Many microbes are fond in nature and help plants to grow

Rhizobacteria found in the soil fixate nitrogen which is required to grow many crops. Furthermore, microorganisms—the cyanobacteria or their DNA in the chloroplasts in plant cells—were the source of most of the free oxygen in the early atmosphere. They also oxidize ammonia (the universal end product of protein metabolism) to nitrate, which is the only nitrogen source used by plants and is therefore essential for production of our plant foods. Microorganisms also are responsible for cellulose hydrolysis in the rumens (first stomach compartments) of cattle, facilitating the production of animal protein for human consumption.

Oxygen production

Cyanobacteria or blue green algae produce oxygen in the ocean and this oxygen production is very necessary for aquatic life survival. Blue-green algae (cyanobacteria) are prokaryotes (that is, their cells have no distinct nucleus). They are very independent nutritionally since they can perform photosynthesis using chlorophyll a. Thus, they can synthesize sugars for energy from

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carbon dioxide using the sun's radiation. They also release oxygen. They can respire aerobically and can fix nitrogen, generating amino acids and protein. They require only water, nitrogen gas, oxygen, carbon dioxide, some minerals, and sunlight. The evidence is that they were on earth 3.2 billion years ago. The cyanobacteria are among the earliest microorganisms and very important even today

Decomposition

Decomposition – Defined as the breakdown of raw organic materials to a finished compost – The fungi invade the organic matter in soils first and are then followed by bacteria. – Without this recycling of inorganic nutrients, primary productivity on the globe would stop.

Q17. Differentiate Between Transcriptomics and Proteomic? 5 Marks

Transcriptomics

The presence of a gene in a genome does not obviously mean that it is expressed to produce mRNA and subsequently a protein. Furthermore, regulation of metabolism often involves alteration of expression of various genes, and therefore it is important to know their expression level in order to understand the physiology of the cell. Before the genome sequences became available, gene expression had to be examined on a one-by-one basis. The techniques used included Northern blot, which involves annealing of fractionated mRNA with the cognate pieces of DNA, and reverse transcriptase– PCR, in which mRNA is used as a template to produce cDNA, which is then amplified by PCR. These methods are still used when we already know the mechanism of regulation in precise detail and when we want to know the expression level of the genes involved. However, when the regulation mechanism is not known entirely, examining the expression levels of only a few selected genes out of many thousands becomes pure guesswork and has led to many wrong conclusions.

Proteomic

Proteomic analysis requires first the separation of thousands of proteins or their fragments and then their identification. It is impossible to identify this many proteins and their fragments by

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using the traditional approaches, such as reactivity with specific antibodies or functional properties. Thus, the advances in mass spectrometry, now almost the universal approach in identification, were the major factor that made proteomics possible. Previously, mass spectrometry, which measures the mass-to-charge ratio of ions in a vacuum required the "hard" ionization methods that fragmented the sample molecule. Mass spectrometry became extremely useful in the study of proteins and peptides when the two "soft ionization" methods – matrix-assisted laser desorption/ionization (MALDI) and electrospray ionization (ESI) were developed. These methods made possible the analysis of (unfragmented) peptides and proteins at high sensitivity and great precision. The significance of these advances may also be seen from the fact that two of the Nobel prizes in chemistry in 2002 were awarded to scientists who contributed to the development of these methods.

Q18. Transgenic Plant? 5 Marks

Methods dependent on microbial biotechnology greatly increase the diversity of genes that can be incorporated into crop plants and dramatically shorten the time required for the production of new varieties of plants. It is now possible to transfer foreign genes into plant cells. Transgenic plants that are viable and fertile can be regenerated from these transformed cells, and the genes that have been introduced into these transgenic plants are as stable as other genes in the plant nuclei and show a normal pattern of inheritance. Transgenic plants are most commonly generated by exploiting a plasmid vector carried by Agrobacterium tumefaciens, a bacterium. Foreign DNA carrying from one to 50 genes can be introduced into plants in this manner, with the donor DNA originating from different plant species, animal cells, or microorganisms. Higher plants have genes whose expression shows precise temporal and spatial regulation in various parts of plants for example, leaves, floral organs, and seeds that appear at specific times during plant development and/or at specific locations, or whose expression is regulated by light. Other plant genes respond to different stimuli, such as plant hormones, nutrients, lack of oxygen (anaerobiosis), heat shock, and wounding. It is therefore possible to insert the control sequence(s) from such genes into transgenic plants to confine the expression of foreign genes to specific organelles or tissues and to determine the initiation and duration of such expression. Microorganisms that live on or within plants can be manipulated to control insect pests and

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fungal disease or to establish new symbioses, such as those between nitrogen-fixing bacteria and plants.

Q20. What is a Vaccine Adjuvant? 3 Marks

An adjuvant is a substance that is formulated as part of a vaccine to enhance its ability to induce protection against infection. The word "adjuvant" comes from the Latin adjuvare and means "to help." Adjuvants help activate the immune system, allowing the antigens—pathogen components that elicit an immune response—in vaccines to induce long-term protective immunity.

Q21. Define Live, attenuated vaccines and give example? 3 Marks

Live, attenuated vaccines contain a version of the living microbe that has been weakened in the lab so it can't cause disease. Because a live, attenuated vaccine is the closest thing to a natural infection, these vaccines are good "teachers" of the immune system.

Example: Vaccines against measles, mumps, and chickenpox.

Q22. Give the economical beneficial of microbial biotechnology in Pakistan? 3 Marks

The Green Revolution ushered in the late 1960s has transformed some countries from a food-grain importer to a self-sufficient one. However, with our burgeoning population having already crossed the 150 million marks, the achievements of the Green Revolution are unlikely to be sufficient. Injurious but common farm practices have damaged the cultivated land through water and wind erosion, compaction, salinization, and water logging. Our forest cover is a sad mismatch of desirable levels and actual ground realities. Over-tillage rapidly depletes prime agricultural lands and today little quality land is available to increase the area under farm production. It is therefore not surprising to learn that Pakistan will need to import an estimated 10-12 million tons of food-grains to meet its basic requirement of food by the year 2050.

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Q23. Write down the name of the steps in the production of vaccine? 5 Marks

Stages of vaccine production

Vaccine production has several stages. Process of vaccine manufacture has the following steps:

- Inactivation – This involves making of the antigen preparation
- Purification – The isolated antigen is purified
- Formulation –The purified antigenis combined with adjuvants, stabilizers and preservatives to form the final vaccine preparation.

Q24. Name the adjuvants used in modern vaccine? 2 Marks

Al(OH)3 – aluminium hydroxide and/or AlPO4 – aluminium Phosphate

Q25. Two Benefits of Probiotic? 2 Marks

They help improve digestion, safeguard the immune system, and keep the body's intestinal flora in balance.

Q26. Matrix-assisted laser desorption? 2 Marks

Soft ionization" methods – matrix-assisted laser desorption/ionization (MALDI) were developed. These methods made possible the analysis of (unfragmented) peptides and proteins at high sensitivity and great precision.

Q27. Is this possible to mark protein array? 2 Marks

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Indeed, this is possible. In one study, 5800 open reading frames in the yeast genome were cloned and expressed as a fusion with GST and a hexa histidine tag. The proteins were purified on glutathione-agarose and attached to glass slides coated with nickel, using the affinity of the hexa histidine tag.

Q28. Define Prebiotic? 2 Marks

Prebiotic was described as "a non-digestible food ingredient that beneficially affects the host by selectively stimulating the growth and/or activity of one or a limited number of bacteria in the colon, and thus improves host health.

Q29. Define Subunit Vaccine and give example. 3 Marks

Instead of the entire microbe, subunit vaccines include only the antigens that best stimulate the immune system. In some cases, these vaccines use epitopes the very specific parts of the antigen that antibodies or T cells recognize and bind to. immunization. Because subunit vaccines contain only the essential antigens and not all the other molecules that make up the microbe.

Example: Plague immunization.

Q30. Define Vaccine. 2 Marks

A vaccine is a biological preparation that improves immunity to a particular disease. A vaccine typically contains an agent that resembles a disease-causing microorganism and is often made from weakened or killed forms of the microbe.

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